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### R-PHY map advance time selection

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#### Abstract

Systems and methods for dynamically adjusting a MAP advance time for a MAP message sent from a CCAP core to an RPD, based on a comparison of at least one measured latency value against at least one threshold.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Application No. 63/313,600 filed Feb. 24, 2022, the content of which is incorporated herein by reference in its entirety.

### BACKGROUND

(1) The subject matter of this application generally relates to distributed access architectures of a hybrid CATV network, and more particularly to R-PHY (remote physical) architectures that distribute the functions of the Cable Modem Termination System into the network.

(2) Although Cable Television (CATV) networks originally delivered content to subscribers over large distances using an exclusively RF transmission system, modern CATV transmission systems have replaced much of the RF transmission path with a more effective optical network, creating a hybrid transmission system where cable content terminates as RF signals over coaxial cables, but is transmitted over the bulk of the distance between the content provider and the subscriber using

optical signals. Specifically, CATV networks include a head end at the content provider for receiving signals representing many channels of content, multiplexing them, and distributing them along a fiber-optic network to one or more nodes, each proximate a group of subscribers. The node then de-multiplexes the received optical signal and converts it to an RF signal so that it can be received by viewers. The system in a head end that provides the video channels to a subscriber typically comprises a plurality of EdgeQAM units operating on different frequency bands that are combined and multiplexed before being output onto the HFC network.

(3) Historically, the head end also included a separate Cable Modem Termination System (CMTS), used to provide high speed data services, such as video, cable Internet, Voice over Internet Protocol, etc. to cable subscribers. Typically, a CMTS will include both Ethernet interfaces (or other more traditional high-speed data interfaces) as well as RF interfaces so that traffic coming from the Internet can be routed (or bridged) through the Ethernet interface, through the CMTS, and then onto the optical RF interfaces that are connected to the cable company's hybrid fiber coax (HFC) system. Downstream traffic is delivered from the CMTS to a cable modem in a subscriber's home, while upstream traffic is delivered from a cable modem in a subscriber's home back to the CMTS. Many modern HFC CATV systems have combined the functionality of the CMTS with the video delivery system (EdgeQAM) in a single platform called the Converged Cable Access Platform (CCAP).

(4) As networks have expanded and head ends have therefore become increasingly congested with equipment, many content providers have recently used distributed architectures to spread the functionality of the CMTS/CCAP throughout the network. This distributed access architecture (DAA) keeps the cable data and video signals in digital format as long as possible, extending the digital signals beyond the CMTS/CCAP deep into the network before converting them to RF. It does so by replacing the analog links between the head end and the access network with a digital fiber (Ethernet/PON) connection.

(5) One such distributed architecture is Remote PHY (R-PHY) distributed access architecture that relocates the physical layer (PHY) of a traditional CCAP by pushing it to the network's fiber nodes. Thus, while the core in the CCAP performs the higher layer MAC layer processing, the R-PHY device in the node converts the downstream data sent by the core from digital to analog to be transmitted on radio frequency, and converts the upstream RF data sent by cable modems from analog to digital format to be transmitted optically to the core.

(6) Once the functionality of the CMTS/CCAP is divided between a Mac layer in, e.g. a CCAP core and various PHY devices throughout the network, however, synchronously coordinating the transmissions between the downstream cable modems and the CCAP core becomes much more difficult. Specifically, to reduce interference among upstream transmissions, the CCAP sends downstream scheduling messages informing each cable modem of upcoming transmission windows assigned to it, and these scheduling messages must be received a sufficient amount of time before such transmission windows. Ordinarily, a CCAP would be able to reliably calculate the transmission delay between when it sends such messages and the time a cable modem receives the messages because the conditions between the CCAP and modem typically do not change rapidly. Such rapid changes do exist in an R-PHY system, however. What is therefore desired are improved systems and methods in an R-PHY architecture for determining the delay between the time a MAP message is sent and the time it is received.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the invention, and to show how the same may be carried into effect, reference will now be made, by way of example, to the accompanying drawings, in which:

- (2) FIGS. 1A and 1B shows exemplary R-PHY architectures where a CCAP core is used to synchronously schedule transmissions to and from a plurality of cable modems.
- (3) FIG. 2 shows one exemplary timing systems for determining a MAP advance time delay
- (4) FIG. 3 shows an alternate timing system for determining a MAP advance time delay.
- (5) FIG. 4 shows an exemplary method according to one embodiment of the present disclosure.
- DETAILED DESCRIPTION

(6) For purposes of the disclosure and the claims, the following terms are defined to as to more easily understand the subject matter described and claimed: Master Clock: a clock that sends timing information to a slave clock for that clock to synchronize its time to that of the master clock. Slave Clock: a clock that receives timing information from a master clock to synchronize its time to that of the master clock. Grandmaster Clock: a clock that only operates as a master clock and is the source of time to the packet network: Boundary Clock: a clock operates as both a slave and a master by having one port in a slave state receiving time from a master clock, and one or more ports in a master state which disseminate timing information to slaves. MAP messages: messages sent by the CCAP containing bandwidth allocation maps (MAP). The MAP contains information that indicates when a cable modem can transmit and for how long. The CCAP needs to send MAP messages ahead of time, so the cable modem will not miss the transmit opportunity. MAP advance time: The amount of time that the CCAP sends the MAP messages ahead of the transmit opportunity of a cable modem. DOCSIS tics: a unit of time expressed in units of mini-slots since CMTS initialization time, where a mini-slot is 6.25  $\mu$ sec.

(7) As already noted, in R-PHY systems, the clocks of the Remote PHY Devices (RPDs) and the CCAP core must be synchronized in both phase and frequency to properly schedule data transfers between network components. FIG. 1A shows an exemplary topology **10** used to synchronize the devices in an R-PHY architecture. Topology **10** may include a CCAP core **12** synchronized with an RPD **13** connected together via a plurality of network switches **14**. The RPD **13** is in turn connected to one or more cable modems **15**. Synchronization is attained by a clock **16** in the core **12**, acting as a grandmaster clock, which sends timing information to a slave clock **17** in the RPD **13**. Those of ordinary skill in the art will appreciate that, although FIG. 1 shows only one RPD **13** connected to the core **12**, many such RPDs may be simultaneously connected to the core **12**, with each RPD having a slave clock **17** receiving timing information from the grandmaster clock **16** in the core. Those of ordinary skill in the art will also appreciate that an alternative timing architecture could include a separate grandmaster clock.

(8) FIG. 1B shows an alternate topology **18**, which differs from the topology **10** in that a separate grandmaster clock **19** is used to synchronize both the clock **16** in the core **12** and the clock **17** in the RPD **13**. In this architecture, the clock **16** operates as a boundary clock where it is a slave to the grandmaster clock **19** but acts as a master clock to the slave clock **17** in the RPD **13**.

(9) Also as already noted, the CCAP core **12** operates as a MAC layer in an R-PHY system and is responsible for creating and sending periodic downstream MAP packets i.e., scheduling messages to the cable modems **15** so as to coordinate upstream transmissions among the network of cable modems **15**. In turn, the cable modems **15** use the received MAP messages to determine when they may each gain access to the upstream channel and transmit packets in the upstream direction. These MAP messages must be received a sufficient amount of time before the transmission windows included in the MAP messages are scheduled to begin. Typically, the CMTS is configured with the maximum time needed for a MAP message to propagate through the HFC network to a cable modem, so that the CMTS can make sure it transmits MAPS early enough to be usable when it arrives at the cable modem. In legacy systems not based on an R-PHY architecture, a CCAP could reliably calculate the maximum transmission delay between when a MAP message is sent and when it will be received by a cable modem because the conditions between the CCAP and a cable modem are mostly known in advance.

(10) In an R-PHY system, however, a variable latency is added in the transmission of a MAP

message because the MAC layer and PHY layer of the CMTS are geographically separated by a packet switched network, and this latency often cannot be predicted by an operator. Thus, when the MAC layer prepares a bandwidth MAP and sends it to the PHY layer, it must determine the extra delay of traversing the packet switched network between the MAC and PHY layers when determining how early to send a Bandwidth MAP packet. Knowledge of the delay from the MAC layer to the PHY layer is crucial to the operation and performance of the system.

(11) FIG. 2 shows a method for determining an appropriate MAP advance time in an R-PHY system. In this method, a CCAP core **20** has a master clock **21** used to synchronize its time to that of a slave clock **23** in an RPD **22**, ensuring that each of the core **20** and the RPD **22** are synchronized to the same DOCSIS ticks since CMTS initialization. Those of ordinary skill in the art will appreciate that the master clock **21** may be a grandmaster clock or may in turn be a boundary clock that is a slave to a separate grandmaster clock.

(12) The CCAP core **20** may utilize a DEPI Latency Measurement (DLM) message exchange protocol, which uses Downstream External PHY Interface (DEPI) messages that serve to measure the latency between the CCAP Core **20** and RPD **22**. Specifically, The CCAP Core **20** sends a DLM request message **24** to the RPD **22**. Before transmitting the DLM request message **24** on the packet switched network, the CCAP Core **20** writes a timestamp in the packet, recording the CCAP Core transmit time of the packet. The units for this timestamp are preferably DOCSIS ticks. When the DLM request **24** arrives at the RPD **22**, the RPD **22** records the time of packet arrival. The RPD **22** then forms a DLM response message **26**, which contains the timestamp from the CCAP Core **22** and a timestamp representing the time of packet arrival at the RPD **22**—both in units of DOCSIS ticks. The RPD **22** then transmits this DLM response to the CCAP Core, which then subtracts the CCAP Core timestamp from the RPD timestamp, and thereby get a resultant value which is the one-way latency **28** from the CCAP Core **20** to the RPD **22** in DOCSIS ticks. This time can be converted to units of seconds. Once the CCAP Core **20** has the latency measurement, it can make estimates of the other delays in the time it takes for the Bandwidth MAP packet to make it from the CCAP Core **20** to the cable modem, and use these estimates along with the measure of the latency from the CCAP Core **20** to the RPD **22** to determine the ideal time to transmit a Bandwidth MAP to have it arrive at a cable modem within tolerance.

(13) The method shown in FIG. 2 may have several disadvantages. First, DLM requires extra dataplane messages to achieve its goal. Second, DLM requires extensive changes on the CCAP Core and RPD to be able to format and process DLM messages and handle message error scenarios. In addition, the CCAP Core needs to track periodically sent messages and outstanding messages. Third, a CCAP Core may be connected to hundreds of RPDs. Periodically sending a DLM request and processing a DLM response from potentially hundreds of RPDs may overwhelm the processing capability of the CCAP Core.

(14) FIG. 3 shows an alternate system and method for determining an appropriate MAP advance time in an R-PHY system, which instead of exchanging separate messages devoted to measuring the packet delay from a CCAP core **30** to an RPD **32**, uses data from the MAP messages themselves to estimate delay and adjust a MAP advance time accordingly. MAP messages include a MAP allocation start time, which is the start time of the temporal window for which the MAP allocates bandwidth. By examining the map allocation start time in a received MAP and comparing it to the time the RPD **32** receives the MAP message, an RPD **32** can determine the actual measured “HFC map lead time,” which is the receipt time relative to the MAP allocation start time. The RPD **32** can record, store, and report this data as needed back to the CCAP core **30**. When determining its MAP advance time, i.e. the time that the CCAP core **30** calculates that it should send a MAP message in advance of the window for which the MAP is applicable, the CCAP core **30** will have made certain assumptions about the latency between the MAC layer and PHY layer, and from those assumptions determine an expected HFC map lead time by which the RPD **32** will have been expected to receive the MAP. The MAC layer device can compare the measured HFC

map lead time that is reported by the PHY layer device, to its own expected HFC map lead time. From this comparison the MAC layer can determine if its assumed latency is correct or not, and can adjust the assumed latency until the expected HFC map lead time matches the measured HFC map lead time.

(15) Specifically, a system may include a CCAP core **30** with a master clock **31** connected via a packet switched network to an RPD **32** having a slave clock **33**. The master clock **31** and the slave clock **33** ensure that each of the core **30** and the RPD **32** are synchronized to the same DOCSIS ticks since CMTS initialization. The CCAP core **30** may construct a MAP message **34** which includes a start time for the window that the MAP allocates bandwidth, recorded as a `start_alloc_time_mini_slots`. The MAP message is transmitted to the RPD **32**, which would ordinarily pass the MAP message downstream onto its RF port. Prior to doing so, however, the RPD **32** parses the MAP message to retrieve the `start_alloc_time_mini_slots`, and records that value along with a time stamp for receipt of the MAP message, `pkt_arrival_time_docsis_tick`.

(16) The `start_alloc_time_mini_slots` value in the Bandwidth MAP packet is expressed in units of mini-slots since CMTS initialization time. The RPD **32** preferably converts this time stamp into units of DOCSIS ticks since CMTS initialization. This can be done by looking at the “mini-slot size” associated with the MAP, which is configured on the RPD **32** by the CCAP Core **30** at RPD initialization. The configured mini-slot size parameter is the size of a single mini-slot in DOCSIS ticks. Thus, the `start_alloc_time_mini_slots` in the Bandwidth MAP packet is multiplied by the mini-slot size to give the resultant alloc start time expressed in DOCSIS ticks which is stored as `map_alloc_start_time_docsis_tick`. The RPD may also estimate the time it takes from recordation of the `pkt_arrival_time_docsis_tick` to the time that the Bandwidth MAP packet is actually transmitted downstream on its RF port. This estimate is fairly constant for the RPD **32**, as the processing time for Bandwidth MAP packets does not vary much over time. This constant is also preferably expressed in units of DOCSIS ticks and may be denoted as `est_processing_time_docsis_ticks`.

(17) A time expressed in units of DOCSIS ticks can be converted to seconds via multiplying by the constant `seconds_per_docsis_tick`, which is equal to  $1/10240000$ . Thus, The RPD can create a message **36** comprising the time in seconds that the RPD **32** actually received the MAP message **34** in advance of the MAP start allocation time via the following formula:

$$\text{hfc\_map\_lead\_time\_secs} = (\text{map\_alloc\_start\_time\_docsis\_tick} - (\text{pkt\_arrival\_time\_docsis\_tick} + \text{est\_processing\_time\_docs\_tick})) * \text{seconds\_per\_docsis\_tick}.$$

The message **36** may then be sent to the CCAP core **30** which may use the measured MAP lead time to dynamically adjust its MAP advance time. In some embodiments, the CCAP core **30** may examine the recorded samples and determine if the measured HFC map lead time is within a tolerance threshold for a portion of the RF plant attached to the reporting RPD **32**. If the HFC map lead time is out of tolerance, the CCAP Core **30** can take a number of actions, including but not limited to (i) alerting the operator via logging or other user interface; (ii) adjusting its internal algorithms for how early to transmit Bandwidth MAPs to the RPD for the purpose of bringing the HFC map lead time back into tolerance; and (iii) collecting and storing data for later analysis to allow for identifying network issues and debugging.

(18) In some embodiments, the RPD **32** may examine a MAP packet periodically and record the HFC map lead time for these packets. Each HFC map lead time is recorded a sample in a database and stored on the RPD **32**. The RPD **32** may in some embodiments store the most recent X samples for retrieval by the CCAP Core **30** or a cable operator. The RPD **32** may provide storage for a number of samples, because the latency of the network between the CCAP Core **30** and the RPD **32** can vary over time based on routing, congestion, and other factors.

(19) In many cases it may not be desirable to frequently change the MAP advance time, as doing so requires processing resources, particularly based on only small changes in latency. Notably, however, some operators use networks that provide a normal path and a redundant path from the

core to the RPD where the redundant path typically is longer, requiring a longer MAP advance time since a MAP advance time that is too short can result in data loss and modem deregistration. However, when using the shorter normal path, a lower MAP advance time helps to increase performance since a shorter MAP advance time leads to better user throughput and lower delays.

(20) Disclosed are novel systems and methods that avoid excessive MAP advance time adjustments, and maintains user throughput while avoiding data loss. Specifically, the present specification discloses systems and methods that utilize a small number of discrete MAP advance time values, and switches between these values based upon a comparison of a measured latency metric against a threshold latency metric. In exemplary embodiment, there may be only two tiers of MAP advance time values and a single threshold latency metric. Other embodiments may employ more tiers of MAP advance time values, and more associated threshold latency metrics.

(21) FIG. 4 shows an exemplary method according to one embodiment of the disclosure. At step **110** a latency metric is selected and at step **112** a counter is set. At step **114** a low MAP advance time is adopted by a CCAP core, such as core **30** and is sent in a downstream MAP message. At step **116** one or more latency metrics Lm are obtained. The latency metric may be obtained in any appropriate manner. In one embodiment, for example, latency from a CCAP core to a remote device may be measured by the CCAP core using Downstream External PHY Interface (DEPI) latency measurements. In other embodiments, latency measurements may be calculated based on Precision Timing Protocol (PTP) message exchanges between a CCAP core and a remote device. These latency measurements may be computed by the CCAP core, or alternately may be computed by a remote device and sent to a CCAP core.

(22) At step **118** the computed latency metric(s) may be compared to one or more thresholds L.sub.s. If the latency metric is greater than the applicable threshold, then a higher MAP time is selected at step **120**. In a preferred embodiment, at optional step **122**, a counter is set from an initial value of zero to a value of “N.” As explained below, this counter is used to ensure that the higher MAP value only decreases after a threshold number of measurements show diminished latency. Thus, in this specific preferred embodiment, a single instance of a latency measurement exceeding a threshold may cause the method to use a higher MAP advance time, but that higher MAP time is not reduced until a number of successive measurements indicate that doing so is warranted. This precaution adds robustness to the method so minimize the number of instances in which a MAP message is not received on time.

(23) After the counter is set at step **122**, a new latency measurement is obtained at step **116** and then again compared to a threshold at step **118**. If the comparison shows that the measured latency is still higher than the threshold, then the higher MAP time is again selected, and the counter remains at the value “N.” (Those of ordinary skill in the art will realize that, if more than two MAP times are used, necessitating a comparison to more than one threshold, then at step **120**, the MAP time may be increased even further).

(24) Returning to step **118**, if the measured latency is less than the applicable threshold, then at step **124** it is determined whether the counter is zero. If it is not, then the counter is decremented by one, and the method returns to step **116** for a new latency measurement, and so forth. The foregoing procedures occur until the counter is decremented to zero, at which point the procedure reverts to step **114** where the MAP time decreases to the lower value.

(25) It will be appreciated that the invention is not restricted to the particular embodiment that has been described, and that variations may be made therein without departing from the scope of the invention as defined in the appended claims, as interpreted in accordance with principles of prevailing law, including the doctrine of equivalents or any other principle that enlarges the enforceable scope of a claim beyond its literal scope. Unless the context indicates otherwise, a reference in a claim to the number of instances of an element, be it a reference to one instance or more than one instance, requires at least the stated number of instances of the element but is not intended to exclude from the scope of the claim a structure or method having more instances of that

element than stated. The word “comprise” or a derivative thereof, when used in a claim, is used in a nonexclusive sense that is not intended to exclude the presence of other elements or steps in a claimed structure or method.

## Claims

1. A method comprising: setting a plurality of predetermined Bandwidth Allocation Map (MAP) advance time values; measuring an amount of latency between a Converged Cable Access Platform (CCAP) core and a remote device transmitting in an upstream direction; selecting one of the plurality of predetermined MAP advance time values based upon the measured amount of latency; and sending a reservation MAP to the remote device having the identified MAP advance time.
2. The method of claim 1 including the step of comparing the measured amount of latency to at least one threshold.
3. The method of claim 2 where the number of thresholds is one fewer than the number of the plurality of predetermined MAP advance time values.
4. The method of claim 1 including the step of modifying a value of a counter based on a comparison of the measured latency to a threshold.
5. The method of claim 4 where the counter is either increased or decreased based on whether the measured latency exceeds or does not exceed the threshold.
6. The method of claim 4 where the counter is decremented when the counter is greater than one and the measured latency is not greater than the threshold.
7. The method of claim 1 where the number of the plurality of predetermined MAP advance times is two.
8. The method of claim 1 where the number of the plurality of predetermined MAP advance times greater than two.
9. The method of claim 1 where a higher-value predetermined MAP time is selected when a signal instance of the measured latency value exceeds a threshold.
10. The method of claim 9 where a lower-value predetermined MAP time is selected after a determination that a plurality of consecutively-measured latency values have not exceeded the threshold.
11. A Converged Cable Access Platform (CCAP) core configured to: communicate with a remote device in a communications network; and determine a latency associated with the transmission of a Bandwidth Allocation Map (MAP) message to the remote device and use the determined latency to select from among a plurality of predetermined MAP advance time values to communicate in the MAP message.
12. The CCAP core of claim 11 configured to compare the determined latency to at least one threshold.
13. The CCAP core of claim 12 where the number of thresholds is one fewer than the number of the plurality of predetermined MAP advance time values.
14. The CCAP core of claim 12 configured to modify a value of a counter based on the comparison.
15. The CCAP core of claim 14 where the counter is either increased or decreased based on whether the determined latency exceeds or does not exceed one or more of the at least one threshold.
16. The CCAP core of claim 14 configured to decrement the counter when the counter is greater than one and the determined latency is not greater than one or more of the at least one threshold.
17. The CCAP core of claim 11 where the number of the plurality of predetermined MAP advance times is two.
18. The CCAP core of claim 11 where the number of the plurality of predetermined MAP advance times greater than two.
19. The CCAP core of claim 11 configured to select a higher-value predetermined MAP time when

a signal instance of the determined latency value exceeds one or more of the at least one threshold.

20. The CCAP core of claim 19 configured to select a lower-value predetermined MAP time after a determination that a plurality of consecutively-measured latency values have not exceeded one or more of the at least one threshold.

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